

Read in the dataset 'SeroStudy.csv'

```
dat<-read.csv("data/SeroStudy.csv")
```

1. What is the range (hint: use `range()` or `min()` and `max()`) and mean age of participants?
2. What proportion of Male participants are seropositive? (hint: you can use `table()` or the tidyverse `group_by()` + `summarise()` functions)
3. What proportion of Female participants are seropositive?
4. Calculate the proportion seropositive by single age (Hint: You can use a `for` loop or the tidyverse `group_by()` + `summarise()` functions)
Plot age vs seropositivity
5. Using the following relationship, plot the expected proportion seropositive if there was a force of infection of 0.05 in a community:

$$P(\text{seropositive})=1-\exp(-\text{theta} * \text{age})$$

Hint:

- Create an object `theta=0.05`
 - Use a vector of ages between 1 and 15: `ages<-1:15`
6. Plot the line from no. 5 on top of the observed proportion seropositive by age – how is the fit?
 7. Try a few different values of theta and find one that works well.
 8. Now lets use a likelihood approach instead. Using the following likelihood function

$$\sum_{i=1} (1-s_i) \cdot (-\text{theta} * \text{age}_i) + s_i \cdot \log(1 - \exp(-\text{theta} * \text{age}_i))$$

Find the likelihood for candidate values of theta between 0 and 0.1 with increments of 0.01

9. Which value of theta is the likelihood highest? How does this compare to what you found in No. 7.

'Stretch' questions:

10. Using this value of theta, calculate the proportion of 30 year old individuals you would expect to be seropositive.
11. Using this value of theta, calculate the probability of being infected for each age between 1 and 60

12. Assuming that there are 10,000 of each age between 1 and 60 (and no one older) – so 10,000 1 year olds, 10,000 2 year olds etc, calculate how many infections you see in total across all age groups in this population
13. Repeat the last question but assume this time there are 20,000 in each age between 1 and 10, 10,000 between 11-50 and no one older (ie a younger population of the same total size)